# Akoizo Original Practice - Form 5 - Answer key

### Division B | Experimental Design | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

### ANSWER KEY - Review only after submitting.

**1. (2 points) Which is quantitative evidence?**

Unit: Claims, evidence, and reasoning for trends and outliers.
Worked-case data: At levels 1,2,3, means are 12,16,20 units, with SD 2 at each level. Write a bounded trend CER.

A. Means increased from 12 to 20 units

B. It looks nice

C. The hypothesis is obviously true

D. The team worked hard

**Correct: A**

**2. (2 points) Randomizing order mainly helps address…**

Unit: Problem, hypothesis, variables, and experimentation.
Worked-case data: All low-level trials occur early and all high-level trials occur late while the instrument drifts upward.

A. Every possible confounder automatically

B. Time-linked systematic differences

C. All measurement uncertainty

D. The need for units

**Correct: B**

**3. (4 points) Explain the evidence supporting your answer, including the calculation or comparison.**

Unit: A clear Division B investigation report.
Worked-case data: 25 cm^2 towel: Absorbed masses: 2, 3, 4 g. 50 cm^2 towel: Absorbed masses: 5, 6, 7 g. Controls: Same towel brand, soak time, and draining method.
Separate investigation observations:
25 cm^2 towel: Absorbed masses: 2, 3, 4 g.
50 cm^2 towel: Absorbed masses: 5, 6, 7 g.
Controls: Same towel brand, soak time, and draining method.

answer: Calculate (2 + 3 + 4)/3 = 3 g. Calculate (5 + 6 + 7)/3 = 6 g. Compare the means while acknowledging the 2 g range in each group. (4 points).

**4. (2 points) High R^2 proves causation?**

Unit: Quantitative data, graphs, and statistics.
Worked-case data: For response values 8, 10, 12, calculate mean, range, and sample SD.

A. No

B. Yes

C. Only with a colored line

D. Only when n = 3

**Correct: A**

**5. (4 points) Name the example independent and dependent variables.**

Unit: The experiment and the prompt packet.
Worked-case data: A prompt supplies a ramp, object, ruler, and timer. Turn “study motion” into a testable question.

answer: Independent: release height. Dependent: transit time between defined marks. (4 points).

**6. (4 points) Why do more repeats not remove a fixed offset?**

Unit: Errors, conclusions, and recommendations.
Worked-case data: Every timer reading is 0.5 s too high. Would ten more repeats eliminate the bias?

answer: The mean retains the systematic shift even if random uncertainty decreases. (4 points).

**7. (4 points) Why separate raw and summary data?**

Unit: Materials, reproducible procedures, and qualitative data.
Worked-case data: A trial has no timer reading because the observer missed the endpoint. What should the table and procedure record?

answer: It permits auditing and recalculation without confusing original observations with derived quantities. (4 points).

**8. (4 points) Why keep raw data?**

Unit: The experiment and the prompt packet.
Worked-case data: A prompt supplies a ramp, object, ruler, and timer. Turn “study motion” into a testable question.

answer: They allow auditing, recalculation, plotting, and transparent handling of failed trials. (4 points).

**9. (4 points) Recommend a targeted improvement for reaction-time variation.**

Unit: Errors, conclusions, and recommendations.
Worked-case data: Every timer reading is 0.5 s too high. Would ten more repeats eliminate the bias?

answer: Use a defined automated endpoint or more consistent timing method, then verify its calibration and applicability. (4 points).

**10. (6 points) Solve this case and explain your reasoning: A trial has no timer reading because the observer missed the endpoint. What should the table and procedure record?**

reasoning: Mark the value missing, not zero.
Record the missed endpoint as the reason and retain the trial identifier.
Repeat under a predefined rule if feasible, documenting the repeat and its relation to the failed trial. (3 points).

conclusion: Transparent missing-data handling preserves the evidence trail. (3 points).

**11. (4 points) Give a targeted follow-up for excessive setup time.**

Unit: Practice, timing, teamwork, and final review.
Worked-case data: A team finishes data collection with little time left and no graph or analysis. Suggest a targeted practice change.

answer: Use a simpler apparatus, pilot feasibility early, coordinate independent tasks, and reserve analysis time in a new practice run. (4 points).

**12. (6 points) Solve this case and explain your reasoning: At levels 1,2,3, means are 12,16,20 units, with SD 2 at each level. Write a bounded trend CER.**

reasoning: Claim: mean response increases across the tested range.
Evidence: means rise by 4 units per level and by 8 overall, with SD about 2.
Reasoning: a proposed increasing mechanism is consistent, but design validity and sample size limit certainty and extrapolation. (3 points).

conclusion: A complete CER quantifies support and states what it does not establish. (3 points).

**13. (2 points) Define “Control condition” in the context of Experimental Design.**

answer: Reference treatment for comparison. (2 points).

**14. (2 points) Define “Bottleneck” in the context of Experimental Design.**

answer: Task limiting overall progress or quality. (2 points).

**15. (2 points) Define “Evidence” in the context of Experimental Design.**

answer: Relevant observations and quantitative results. (2 points).

**16. (2 points) Define “Protocol deviation” in the context of Experimental Design.**

answer: Departure from the stated procedure. (2 points).

**17. (2 points) Define “Pilot” in the context of Experimental Design.**

answer: Preliminary feasibility check before the final protocol. (2 points).

**18. (2 points) Define “Audit trail” in the context of Experimental Design.**

answer: Traceable record from original observations to reported results. (2 points).